

Expanded Aditya Meeting Master Guide & Layman Playbook

Comprehensive Talking Points, Layman Homology Analogies & Agentic Swarm Demos

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 26, 2026

1. Contact Profiles & Dual-Coverage Context

Aditya Singh (Google Agentic AI Systems Architect): Specializes in agentic AI platforms, GCP scaling, and product impact. Focus on our autonomous swarm orchestrators (`mera-kmpa-swarm-framework`).

Aditya Kumar Singh (S&P; Global Data Analyst): Specializes in index data, financial economics, and GICS sector analytics. Focus on our persistent topology risk models.

2. Layman Analogies for Swarm Persistent Homology (beta_0, beta_1, beta_2)

Topological Invariant	Technical Definition	Crystal-Clear Layman Analogy	Financial & Swarm Impact
beta_0 (Betti-0)	Connected Components	Islands in an Archipelago: Measures how many isolated asset clusters exist independently.	In a crash, isolated islands collapse into 1 mega-island (correlation spikes to 1.0).
beta_1 (Betti-1)	1-Dimensional Loops	Capital Swirls & Whirlpools: Detects circular money flow between assets (Asset A -> B -> C -> A).	Detects liquidity loops and momentum rotation before price surges occur.
beta_2 (Betti-2)	2-Dimensional Cavities	Hollow Balloons / Air Bubbles: Identifies empty voids inside multi-asset data space.	Warns of liquidity voids where price orderbooks vanish, causing flash crashes.

3. Step-by-Step Conversational Agenda (15-Minute Playbook)

[0:00 - 3:00] Warm-Up: 'Hi Aditya, great to connect! I saw your work in agentic AI systems / financial economics. Excited to swap ideas on how non-stationary regime shifts impact complex systems.'

[3:00 - 8:00] Layman Core Concept: Explain the 3 Betti numbers using the Island (beta_0), Whirlpool (beta_1), and Balloon (beta_2) analogies.

[8:00 - 12:00] Live Performance Demo: Highlight WorldQuant BRAIN Sharpe 2.76 / 1.76 Fitness monster alpha results and 0.88 Sharpe Numerai swarm models.

[12:00 - 15:00] Next Steps: Invite collaboration on publishing updated pre-print papers with Stanford Emeritus Prof. Gunnar Carlsson and exploring Google Cloud Agentic AI integrations.